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METHODOLOGY ARTICLE

Exploring a graph theory based algorithm for automated identification and characterization of large mesoscale convective systems in satellite datasets

Kim Whitehall · Chris A. Mattmann · Gregory Jenkins · Mugizi Rwebangira · Belay Demoz · Duane Waliser · Jinwon Kim · Cameron Goodale · Andrew Hart · Paul Ramirez · Michael J. Joyce · Maziyar Boustani · Paul Zimdars · Paul Loikith · Huikyo Lee

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Abstract Mesoscale convective systems are high impact convectively driven weather systems that contribute large amounts to the precipitation daily and monthly totals at various locations globally. As such, an understanding of the lifecycle, characteristics, frequency and seasonality of these convective features is important for several sectors and studies in climate studies, agricultural and hydrological studies, and disaster management. This study explores the applicability of graph theory to creating a fully automated algorithm for identifying mesoscale convective systems and determining their precipitation characteristics from satellite datasets. Our results show that applying graph theory to this problem allows for the identification of features from infrared satellite data and the seamlessly identification in a precipitation rate satellite-based dataset, while innately handling the inherent complexity and non-linearity of mesoscale convective systems.

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K. Whitehall · G. Jenkins · M. Rwebangira · B. Demoz
Howard University, Washington, DC, USA

K. Whitehall · C. A. Mattmann (✉) · D. Waliser · C. Goodale · A. Hart · P. Ramirez · M. J. Joyce · M. Boustani · P. Zimdars · P. Loikith · H. Lee
Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA
e-mail: mattmann@usc.edu

C. A. Mattmann
University of Southern California, Los Angeles, CA, USA

D. Waliser · J. Kim
UCLA Joint Institute for Regional and Earth System Science Engineering, Los Angeles, CA, USA

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Introduction

Mesoscale convective systems (MCSs) are features of organized thunderstorms that vary in size and persists for hours. Mesoscale convective complexes (MCCs) are a subset of large and well-organized mesoscale convective systems (MCSs). MCCs are convectively driven weather systems that are elliptically shaped (eccentricity ~0.7), last approximately 11.5 h and contribute large rainfall amounts to the daily and monthly totals (Maddox 1980; Laing and Fritsch 1997). Many weather-related studies have focused on the smaller-scaled MCSs e.g. cloud clusters (Fig. 1), but their variability is strongly related to the large-scale meteorological conditions and the larger scale mesoscale interactions (Houze et al. 1990). Thus, studies related to the larger scaled MCSs are merited to add to the understanding of convection, and the processes that lead to convection organization at the various meteorologically defined meso-spatial and temporal scales. Approximately 400 MCCs occur annually over locations on the globe between 60°N and 60°S, of which approximately 66% occur in the Northern Hemisphere, which contributes to significant precipitation totals (Laing and Fritsch 1997). MCCs are related to regional precipitation characteristics such as the timing and amounts of seasonal precipitation as well as hydrologic extremes such as flooding. Thus, understanding of the lifecycle, characteristics, frequency and seasonality of MCCs (and other categories of MCSs) is important for several studies such as hydrological studies, climate studies and disaster management.

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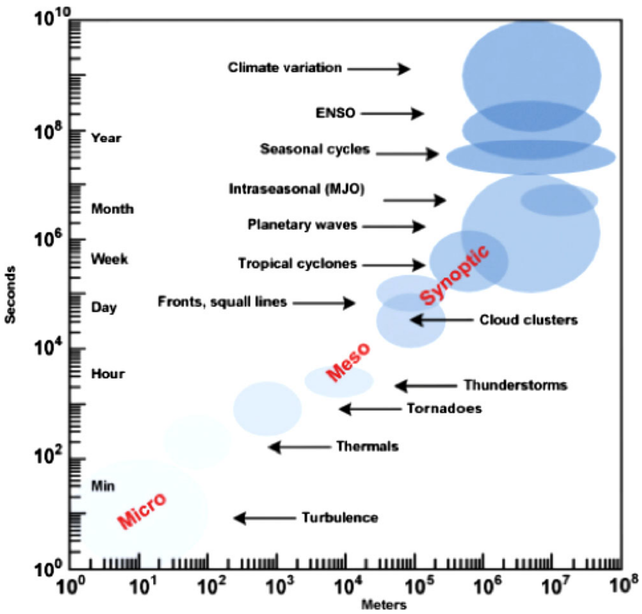


Fig. 1 A schematic of the range of MCSs regarding their spatial and temporal scales, using selected MCS features. Adopted from Laing and Evans 2011

Identifying MCCs from various types of meteorological data is crucial for studying their characteristics and effects. Though some studies identifying MCC precipitation characteristics exist for a particular event or a season of events (e.g. Blamey and Reason 2012; Goyens *et al.* 2011), comprehensive research related to studying the seasonality of MCCs has been limited by methods of dealing with the big data inherent of identifying them in automated methods and other datasets. MCC (and MCS) identification utilizes a spatial-temporal criterion that requires high-resolution (30 min to 3 hourly) geostationary weather satellite data (Maddox, 1980). Initially, the methods for identification were fully manual and required both infrared (IR) images that supplied the cloud-top brightness temperature (BT), and visible (VIS) images that provided information on cloud characteristics e.g. (Velasco and Fritsch 1987; García-Herrera *et al.* 2005). These past studies identified the characteristics of the MCC, their cold cloud shield areas and temperatures, their duration, and their lifecycle. The time-consuming manual method evolved into semi-automatic methods that incorporate an automatic implementation using the IR data and a human factor to corroborate the automatic implementation using VIS images, for example, (Laing and Fritsch 1993; Blamey and Reason 2012), and automated implementations that do not require VIS imagery (Carvalho and Jones 2001; Schröder *et al.* 2009; Vila *et al.* 2008).

The studies that implemented semi-automatic and automatic methods e.g. Machado *et al.* 1998; Machado and Laurent 2004; Mathon and Laurent 2001; Goyens *et al.* 2011; Arnaud *et al.* 1992 added to the knowledge base of the characteristics of MCCs according to location, as well as started to explore the precipitation characteristics of MCCs. The existing and generally utilized semi-automatic and automated methods are based on forward and backward in time algorithms e.g. (Carvalho and Jones 2001; Schröder *et al.* 2009), that tend to be inefficient in computer resources, but more importantly, often falter when evolving cloud systems that may merge or split over time - thus these methods still require a human intervention. More recently, a fully automated method - the Tracking of Organized Convection (TOOCAN) algorithm has been introduced for identifying and tracking MCSs in meteorological applications (Fiolleau & Roca, 2013). The objective of the TOOCAN algorithm is to identify MCSs by associating areas of coldest BTs (convective seeds/cores) in sequential geostationary infrared satellite data through clustering methods in a spatio-temporal domain. The clustering method identifies the convective seeds (areas of deepest convection) according to a BT maximum threshold, then iteratively over temperature ranges (e.g. 1 K variation) track the MCS from this radial point outwards in a 3-dimensional space (time, latitude and longitude) as a MCS. The TOOCAN algorithm addresses the concerns of previous MCSs tracking methods and adds layers of analysis to meteorological applications involving MCSs, but intuitively the algorithm does not appear efficiently scalable to long-term datasets. None of the existing methods for identification of MCSs in infrared datasets allows for the features to be compared with other (gridded) datasets without human intervention.

Graph theory has been marginally used in the atmospheric sciences for now casting (predicting the details of the changing atmosphere for a period between 0 to 6 h over a small area, e.g. a city) and forecasting MCSs, specifically thunderstorms with radar and satellite data. This implementation of graph theory considered the relationship between atmospheric variables at a given time (Chaudhuri and Middey 2009; Chaudhuri and Middey 2011), or the spatial-temporal analysis of cloud volumes through combinatorial optimization of graph theory (Mukherjee and Acton 2002; Dixon and Wiener 1993).

This study explores the feasibility of utilizing graph theory to creating a fully automated method for identifying MCCs (and their MCSs construct) in high resolution geostationary satellite infrared datasets and provide precipitation characteristics. More specifically, in this explorative study, we aim create a standalone algorithm that can identify and characterize large MCSs, specifically MCCs in gridded infrared and precipitation rate satellite data, that is transferable to long-term records of data, and can be implemented within (graph-based) databases of satellite data.

Datasets

The infrared (IR) brightness temperature geostationary satellite data used in this study are the National Centers for

Environmental Prediction / Climate Prediction Center (NCEP/CPC) 4 km Global (60 N – 60S) IR Dataset (also known as MERG – merged – dataset). The CPC, NCEP and the National Weather Service (NWS) created this dataset from IR data collected by different sensors sampling within the 10.2 and 12.5 μm spectral bands on various geostationary satellite platforms. The data are representative of readings from the top of the atmosphere and have been corrected for errors such as the "zenith angle dependence". The data itself is the brightness temperature (BT), which is the temperature equivalent of the radiation emitting from the cloud top (acting as black body) at a given wavelength, and can be used to infer information about the altitude of the cloud and its thickness - the higher a cloud is from the surface and the thicker the layer is, the colder its BT. In preparing the MERG dataset for the public, the aforementioned centers regridded the data such that it exists on a rectangular latitude-longitude 0.036° (~4 km) grid at half-hourly intervals. For this study, hourly MERG data (on the hour data i.e. ":00") will be utilized.

The precipitation satellite data used in this study are from the Tropical Rainfall Measuring Mission (TRMM) – a mission between National Aeronautics and Space Administration (NASA) and the Japan Aerospace Exploration Agency (JAXA) – and provide the distribution of rainfall within the Tropics through combining data from various satellite-based instruments. The 3-hourly TRMM 3B42 version 7 dataset is produced from the multiple TRMM sensors and IR-precipitation estimates from sensors on other satellite platforms, and in the absence of a dense network of surface gauges is best for this analysis in the region (Goyens *et al.* 2011). The TRMM 3B42 version 7 data are available at a 0.25° (~25 km) resolution on a 3-hourly, daily or monthly temporal resolution. For this study, the TRMM 3B42 version 7 3-hourly data that contains equivalent precipitation rate data in mm hr^{-1} for each hour within the 3-hour period will be utilized.

Both datasets were accessed via the NASA Goddard Earth Sciences Data and Information Services Center (GES DISC) Mirador system (http://mirador.gsfc.nasa.gov/) in the Network Common Data Form (netCDF) format.

Methodology and review of existing methods

Automated methods for identifying mesoscale convective systems are usually based on forward and / or backward algorithm that executes in two stages: (1) a cloud detection or cloud masking stage, and (2) a tracking or evolution stage (e.g. Carvalho and Jones 2001; Schröder *et al.* 2009).

The cloud detection or cloud-masking stage involves identifying the cloud areas of interest from a full IR satellite dataset at time increments via temperature, and area or volume restrictions. These regions of interest are referred to here as cloud elements (CEs). The CE criterion varies according to the particular MCS and the location study. Previous studies identified the temperature ranges associated with convective clouds in large MCSs (including MCCs) to range from a 255 K temperature threshold to identify large-scale areas, to a 195 K temperature threshold to identify the (usually smaller) convectively most active areas of the system (Machado *et al.* 1998; Fiolleau and Roca 2013; Duvel 1989; Desbois *et al.* 1988; Mapes and Houze 1993). Temperature thresholds may have limitations regarding the identification of the convective portion of cloud. The area cutoff criterion ranges from 2,400 km^2 to 5,000 km^2 for large MCSs such as mesoscale convective complexes, where smaller area cut-offs tend to produce too many cloud elements that are not of consequence to the MCS identification (Vila *et al.* 2008; Machado and Laurent 2004; Mathon and Laurent 2001).

The tracking or evolution stage observes the life cycle of the mesoscale convective system through correlating the identified areas of interest from the cloud detection stage. The methods existing within the literature, include maximum spatial correlations (Carvalho and Jones, 2001), maximum overlap between identified convective areas (Schröder *et al.* 2009), a Lagrangian approach based on projected centroid location (Johnson *et al.* 1998), propagation speed of criterion (Woodley *et al.* 1980), maximum overlap area (Laurent *et al.* 1998), a global cost function to enforce shape and path characteristics (Chaudhuri and Middey 2011), and greedy optimization of overlap areas (Dixon and Wiener, 1993). In the area-overlap method, CEs are considered to be correlated if there is greater than 50 % or in excess of 10,000 km^2 area spatial overlap between successive images (Williams and Houze 1987). The percentage overlap varies according to the temporal resolution of the data and the smallest CE size being considered (Schröder *et al.* 2009; Goyens *et al.* 2011). Though most common method used in automated tracking algorithms is the area-overlapping method, previous implementations identified a problem relating to spontaneous splitting or merging of cloud areas though they belonged to the same system (Mathon and Laurent 2001). We anticipate that using graph theory in this problem will address that concern.

The algorithm

The algorithm developed for this study is coined to as the "Grab 'em, Tag 'em, Graph 'em"(GTG) algorithm. The GTG algorithm suggested here is based on the science of the aforementioned methods surrounding the science of mesoscale convective system (MCS) identification, but utilized graph theory. The graph theory implementation involves using graphs as objects along with some graph traversal methods. Fig. 2a provides a schematic of the algorithm that works by first identifying areas of interest in the gridded IR satellite dataset to create a graph, then searching the graph for the type

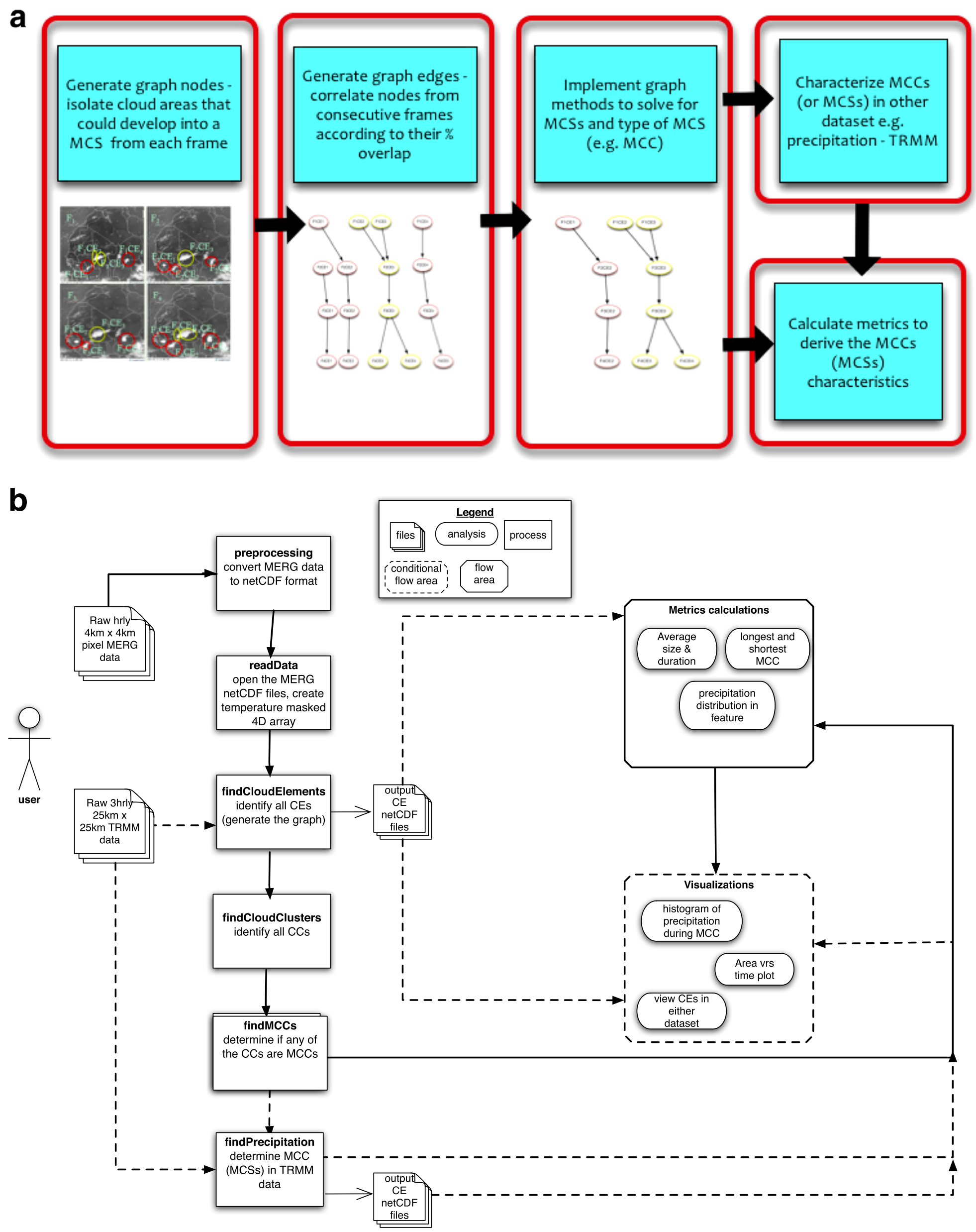


Fig. 2 **a** A schematic of the graph theory algorithm for detection and tracking of MCSs **b** The workflow of the "Grab 'em, Tag 'em, Graph 'em" (GTG) algorithm

of MCS, in this case MCCs. The GTG algorithm implementation is Python based and uses the numerical (NumPy http://www.numpy.org/) and scientific (SciPy http://scipy.org/) libraries in Python (Oliphant, 2007), as well as the graph object and graph methods from the Networkx Package (http://networkx.github.io/, Hagberg *et al.*, 2008). Fig. 2b provides the general workflow of the GTG algorithm.

In this study, each time (image) of IR data is referred to as a frame. Each frame, F_t, is a function of latitude, longitude and brightness temperature (BT), where t =1 h for this study.

$$F_t = \{lat, lon, BT\}$$

Within each frame, there exists a number of CEs. The properties of a CE in the region of interest are defined by a BT and area criteria where the warmest temperature considered for CE identification is 241 K as for deep convection in the Tropics, a buoyant parcel would most likely originate below the 700-hPa level that corresponds to this brightness temperature (Machado *et al.* 1998). Furthermore, Maddox (1980) identified the warmest temperature an MCC cloud shield of as 241 K.

An area cutoff of 2,400 km^2, the smaller value of the range in previous studies, is considered because it allows for further analysis of large MCS evolution (especially in the initiation and decaying stages) and given the graph implementation significant computational resources or human intervention will not be required as in other methods. But areas smaller than 2,400 km^2 are captured in the detection stage where the temperature range within the potential CE correlates it with being a convective element i.e. the convective fraction (cf) is at most 0.9. We consider this other criteria following discussions in the literature about the microphysical characterization of MCS features in the area of interest and previous temperature threshold implementations (Bouniol *et al.* 2010, Machado *et al.* 1993, Mapes and Houze 1993, Goyens *et al.* 2011):

Brightness temperature, BT≤241 K

1. Area, A≥2,400 km^2 or A<2,400 km^2 and (BT minimum in the area/ BT maximum in the area)>=convective fraction, where cf≤0.9
2. Area, A ≥ 2,400 km^2 or A 2,400 km^2 and (BT minimum in the area/ BT maximum in the area) >= convective fraction, where cf ≤ 0.9

$$CE[lat,lon,BT]_t \in \left\{ \begin{array}{l} F_t[lat,lon,BT] \,\middle|\, \left(F_t[lat,lon,BT] \le 241K \wedge \left(\left(F_t[lat,lon,BT] \ge 2400\ km^2\right)\right.\right. \\ \left.\left.\vee\left(\left(F_t[lat,lon,BT] \le 2400\ km^2\right) \wedge \left(\dfrac{F_t[lat,lon,BT_{min}]}{F_t[lat,lon,BT_{max}]} \ge 0.9\right)\right)\right)\right) \end{array} \right\}$$

And,

$$F_t = CE[lat,lon,BT]_{t,n} \in \left\{ n \,\middle|\, CE[lat,lon,BT]_t \right\}$$

As the IR data is a gridded dataset, the multidimensional array image-processing methods within the SciPy package (scipy.ndimage) are implemented to facilitate the CE search criteria as outlined above as the segmentation in the ndimage packages allows for separating objects of interest from a background through probability intensity thresholding. In the GTG algorithm, the geospatial areas of interest identified from the IR dataset, represent the graph nodes. Using graph theory nomenclature, during the cloud detection stage, a directed graph, *G*, is created where the graph nodes, *V (G),* are the CEs for each frame.

Although each CE exists at a discrete time, there is correlation between them over time that represents the MCS life cycle. The GTG utilizes the area-overlapping method (Williams and Houze 1987; Arnaud *et al.* 1992). In the area-overlapping method, the percentage and area overlap between consecutive CEs is considered where the chosen percentage overlap varies according to the CE size, its speed, and the temporal resolution of the data. CEs in consecutive frames that are correlated via the percentage or area overlap are represented by the graph edges. The edges of the graph, *E (G)*, are weighted according to percentage overlap between the geospatial parameters between CEs in F_t and $F_{t+\Delta t}$ and are directed from a parent node in F_t to a child node in $F_{t+\Delta t}$. In the graph construct, it is desired to have minimum weighted edges associated with maximum correlation between nodes. Further, in this experiment the minimum CE size is 2,400 km^2 and MCCs in West Africa have been shown to propagate at speeds between 12–15 ms^{-1} (Laurent *et al.* 1998). As such, the minimum weight of an edge occurs where the percentage overlap between two nodes is at least 66 %, whereas the maximum weight occurs for a 50 % overlap or in excess of 10,000 km^2 (Arnaud *et al.* 1992; Williams and Houze 1987). Thus, we define a weighting function for the edges, $W_{lat,lon}$ such that

$W_{lat,lon}$=($F_tCE_n \cap F_{t+\Delta t}CE_n$) (min weight =66 % area overlap, max weight =50 % area overlap or area overlap≥ 10,000 km^2).

Within the graph construct, the collection of CEs and edges is referred to as a cloud cluster (CC). A CC can have a minimum duration of two frames, and maximum of *m* frames, where *m* is number of frames being considered. As such,

$$CC_n = CE_{n,t} + CE_{n,(t+\Delta t)} + CE_{n,(t+2\Delta t)} + \dots + CE_{n,(t+m\Delta t)}$$

Hence CCs do not necessary involve only one CE from a given frame. From observation (e.g. Mathon and Laurent 2001; Goyens *et al.* 2011), as a CC evolves, CEs from different frames can participate in merging, splitting, growth, decay or maintenance. At this point, the graph depicts all the CCs found within the time period analyzed, where a CC is a subgraph of G. An optional mesoscale convective system (MCS) search can be conducted on each CC to determine the core nodes of the system. This is conducted via a (deepest) Dijkstra shortest-path (Dijkstra 1959) search that is implemented on each subgraph. The Dijkstra shortest-path is a graph algorithm that identifies the minimum cost (the edges define the cost), or the shortest path to traverse from one node of the graph to another. Practical applications of Dijkstra's shortest-path algorithm include finding the shortest route between locations. By adding the deepest (longest) path criteria on to the Dijkstra's shortest-path to the GTG algorithm, the furthest evolution of a cloud element (CE) can be tracked. Furthermore, a minimum length restriction of 3 h is imposed to discard keeping any features that will not meet the minimum duration requirements of large organized MCSs (Goyens *et al.*, 2011). It is assumed that if there exists merging and splitting within a CC, there must be the minimum overlap between consecutive CEs (i.e. weighted Dijkstra shortest-path search). At the end of this step, the returned path indicates the MCS defined as the greatest connectivity between CEs from the beginning of the CC's existence to its end.

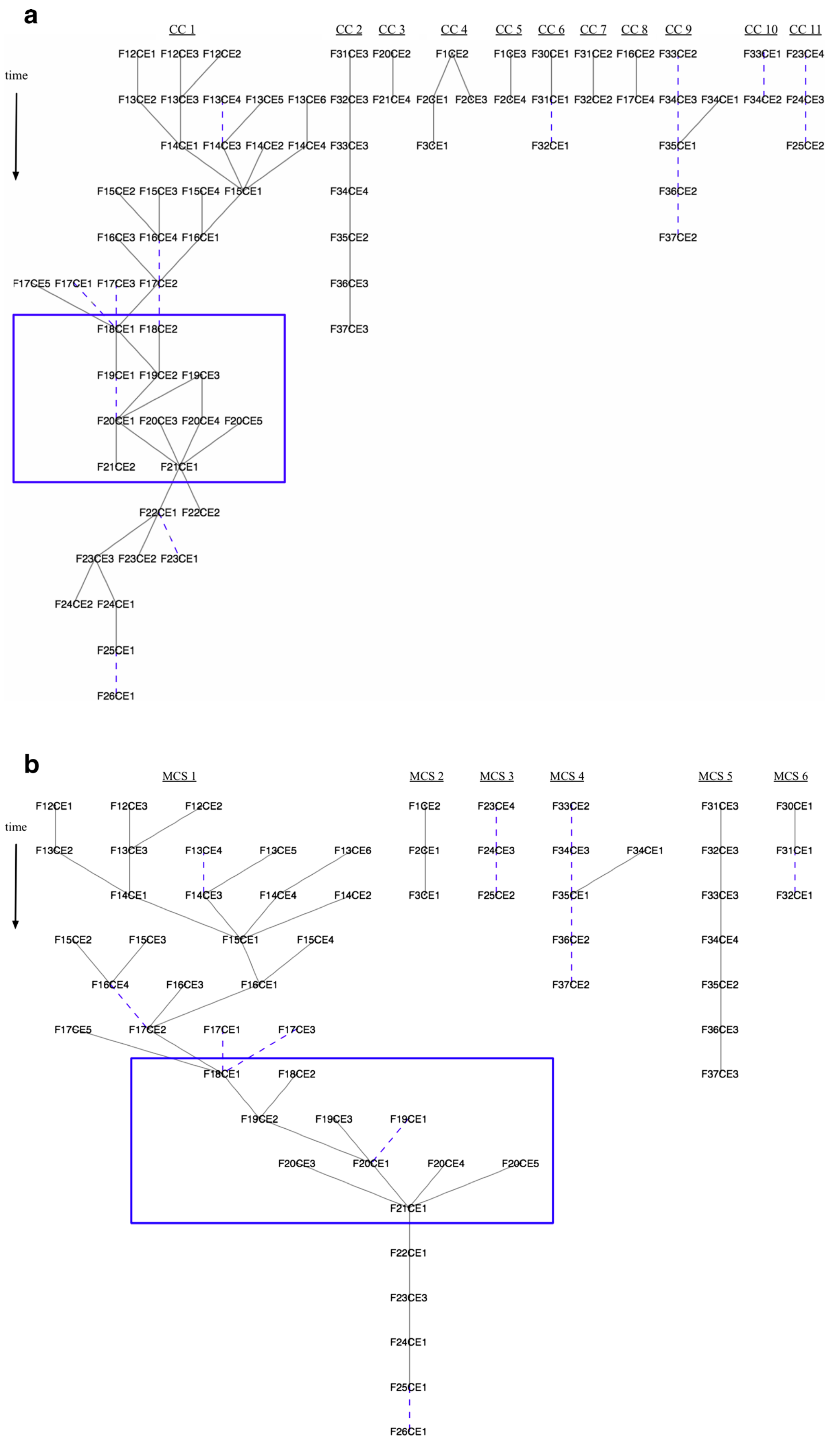
To identify MCCs, each subgraph (CC or MCS) is searched to determine if it meets the criteria. To accomplish this each subgraph is traversed to determine if the MCC criterion for West Africa following Laurent *et al.* (1998) criteria in Table 1 is met, as well as each CE's (node's) contribution. A depth first –iteratively deepening (DFID) graph traversal method is leveraged for this stage (Korf 1985). The DFID is a graph traversal algorithm that addresses a way to visit all the nodes in a graph in a particular order, while checking / or updating the nodes along the way. The DFID traversal specifically accounts for the uniqueness of our subgraphs with respect to: (1) their node connectivity (specifically, a node does not have to be connected a node from the previous frame i.e. a new CE; (2) their complexity, specifically, nodes can form from merges of CEs from the previous frame or splits; and (3) their MCC contribution, specifically, only a subset of consecutive nodes within a subgraph may actually be the MCC. At the end of this step, the returned nodes indicate the CEs that meet the West African MCC criteria. In the case studies presented here the MCSs search was conducted followed by a MCC search using the identified MCSs as input as the purpose is to identify the core features.

The rainfall characteristics associated with the identified IR areas are determined from the TRMM dataset. The GTG algorithm automatically regrids the IR and rainfall rate datasets to common spatial grid and accesses the appropriate geo-spatial data rainfall data for the corresponding IR area through existing functionality from the Apache Open Climate Workbench (Apache OCW http://climate.incubator.apache.org/) project. The Apache OCW is a graduated Apache Software Foundation Top Level Project that contains functionality of data extraction, manipulation e.g. regridding, metrics calculations and visualization, for big datasets in the earth sciences such as satellite and model data in an object-oriented fashion allowing persons to utilize what they require to build their own applications within the project's overarching objective of facilitating climate model evaluations to utilize satellite observations.

Fig. 3 **a** The full graph representing all the nodes / cloud elements and cloud clusters identified during the 36-h period (number of nodes is 76, number of edges are 68). The text (*graph nodes*) represent the cloud elements (CEs) identified. The text indicates the frame number and the cloud element number for that frame. The black lines (*graph edges*) indicate an area overlap ≥65 % or more between CEs, the blue dashed lines (*graph edges*) indicates an area overlap between 65 % and 50 %, whereas the yellow dashed lines (*graph edges*) indicates an area where there was an area overlap≥10,000 km^2. The cloud clusters, CCs (*subgraphs*) are identified with headings CC 1 through to CC 11. The blue box indicates the identified CEs between 11 September 2006 17:00:00 UTC (F18) and 20:00:00 UTC (F21) inclusive. **b** The sub graphs representing the large mesoscale convective systems (MCSs) identified during the 36-h period (number of nodes is 60, number of edges is 54). The text (*graph nodes*) represent the cloud elements (CEs) identified. The text indicates the frame number and the cloud element number for that frame. The black lines (*graph edges*) indicate an area overlap ≥65 % or more between CEs, the blue dashed lines (*graph edges*) indicates an area overlap between 65 % and 50 %, whereas the yellow dashed lines (*graph edge*) indicates an area where there was an area overlap≥10,000 km^2. The cloud clusters, CCs (*subgraphs*) are identified with headings MCS 1 through to MCS 6. The blue box indicates the identified CEs between 11 September 2006 17:00:00 UTC (F18) and 20:00:00 UTC (F21) inclusive. **c** Cloud detection and tracking of MCS 1 from 11 Sep 2006 17:00:00 UTC to 11 Sep 2006 21:00:00 UTC over Niamey, Niger. The first column illustrates the original infrared satellite images, the second column outlines the cloud elements detected by the algorithm presented. The arrows indicate the connections between the CEs after the Dijkstra's shortest-path search (*solid arrows*). The dashed arrow indicates an edge that not a part of the shortest-path found

Table 1 The criteria used for determining MCSs and MCCs in West Africa. The MCS criterion in general follows that of Vila et al. 2008. The MCC criterion follows Laurent et al. 1998, except those researchers did not use a shape criterion. Following the original definition of MCC (Maddox, 1980), we include a shape criterion in our definition that is relevant to the region (Mathon & Laurent, 2001

PHYSICAL CHARACTERISTICS FOR MCS FROM IR DATA	
SPATIAL RESOLUTION	An area 2,400 km^2 where the brightness temperature <241 K or a smaller area where there exists a convective core such there is at least 10 K brightness temperature range
DURATION	≥ 3 h
PHYSICAL CHARACTERISTICS FOR MCC FROM IR DATA	
SPATIAL RESOLUTION	A- Cloud shield with continuously low IR temperature≤−40 °C must have an area≥30,000 km^2
DURATION	A and B must be met for a period≥6 h and≤24 h 24 h - there is no set maximum time in the literature however, it is recognized that if such a feature persists for more than 24 h, it falls within the synoptic scale
MAXIMUM EXTENT	Contiguous cold cloud shield (IR temperature≤- 32 °C) reaches maximum size
SHAPE	Eccentricity (minor axis / major axis)≥0.5 at time of maximum extent at−40 °C


a
CC 1 CC 2 CC 3 CC 4 CC 5 CC 6 CC 7 CC 8 CC 9 CC 10 CC 11
time
F12CE1 F12CE3 F12CE2 F31CE3 F20CE2 F1CE2 F1CE3 F30CE1 F31CE2 F16CE2 F33CE2 F33CE1 F23CE4
F13CE2 F13CE3 F13CE4 F13CE5 F13CE6 F32CE3 F21CE4 F2CE1 F2CE3 F2CE4 F31CE1 F32CE2 F17CE4 F34CE3 F34CE1 F34CE2 F24CE3
F14CE1 F14CE3 F14CE2 F14CE4 F33CE3 F3CE1 F32CE1 F35CE1 F25CE2
F15CE2 F15CE3 F15CE4 F15CE1 F34CE4 F36CE2
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F20CE1 F20CE3 F20CE4 F20CE5
F21CE2 F21CE1
F22CE1 F22CE2
F23CE3 F23CE2 F23CE1
F24CE2 F24CE1
F25CE1
F26CE1
b
MCS 1 MCS 2 MCS 3 MCS 4 MCS 5 MCS 6
time
F12CE1 F12CE3 F12CE2 F1CE2 F23CE4 F33CE2 F31CE3 F30CE1
F13CE2 F13CE3 F13CE4 F13CE5 F13CE6 F2CE1 F24CE3 F34CE3 F34CE1 F32CE3 F31CE1
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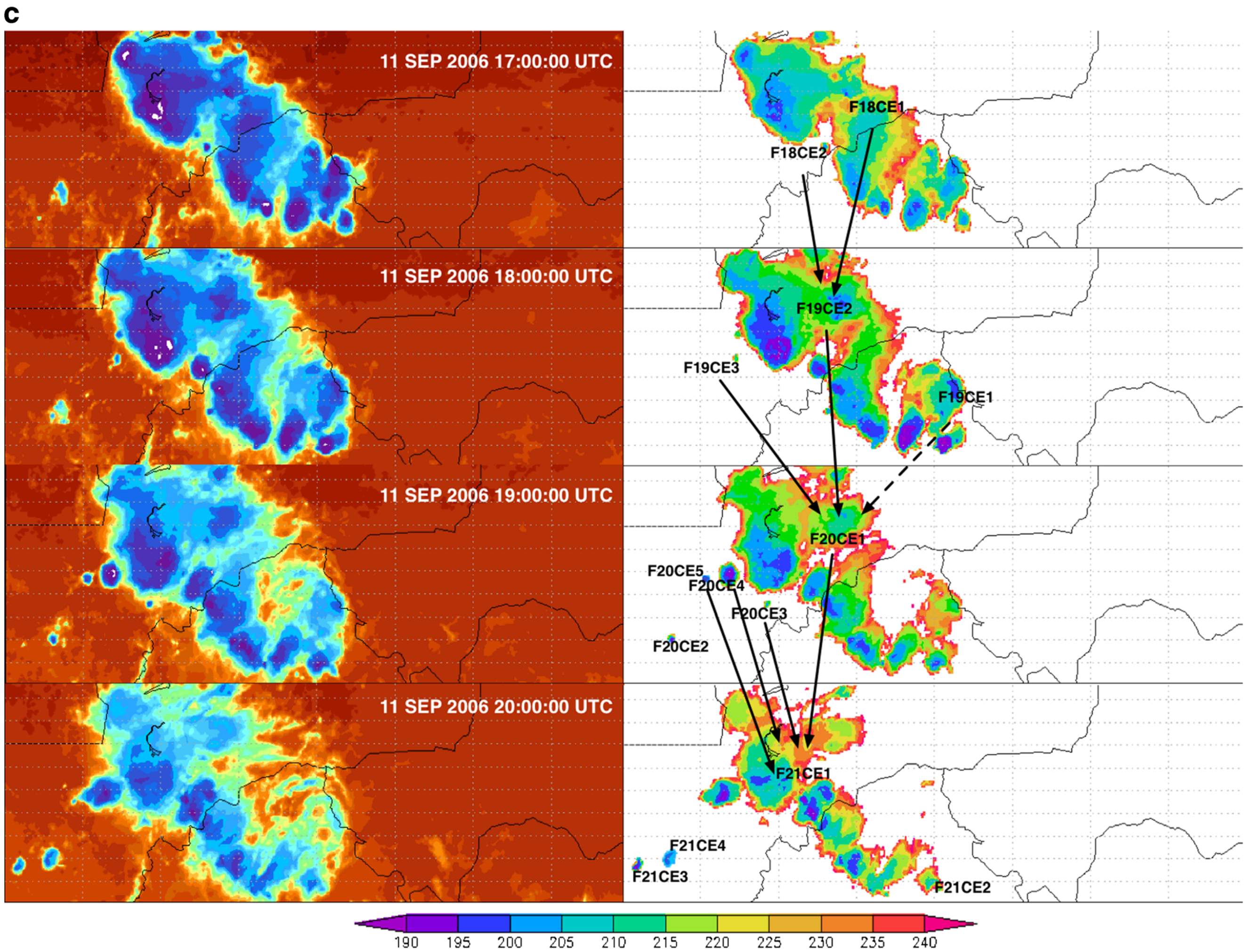

Fig. 3 (continued)

First results and discussion

Case study: Tracking a MCS over Niamey, Niger

The area over Niamey, Niger (12°N - 17°N, 8°E - 8°W) is examined for the 36 h period between 11 September 2006 00:00:00 UTC to 12 September 2006 12:00:00 UTC using the infrared brightness temperature data from the MERG dataset and the 3-hourly TRMMv7 precipitation rate dataset. This location and time period was chosen as other researchers have analyzed the feature using varying identification techniques – the Tracking of Organized Convection (TOOCAN) algorithm (Fiolleau & Roca, 2013) and the area-overlapping method (Mathon & Laurent, 2001). The area-overlapping method (Mathon & Laurent, 2001) results for the location and time were presented in Fiolleau and Roca 2013 study.

The GTG algorithm created a dense graph with of 76 cloud elements (CEs) – areas of interest for MCS development – and 68 edges at varying weights connecting these CEs within this period for the outlined domain (Fig. 3a). Eleven (11) cloud clusters (CCs) – CEs correlated by area overlap over time – are identified. The complexities of these CCs vary. Some CCs show a simple growth and decay pattern, such as a CE appearing at 11 September 2006 00:00:00 UTC (F1CE3) and continued until 01:00:00 UTC (F2CE4) (CC_5 in top middle portion of Fig. 3a), but this feature fully decayed by 02:00:00 UTC (F3 does not exist). Likewise, complicated growth patterns exist between frames involving merging of CEs within the same frame in the subsequent frame. At 11:00:00 UTC (F12), the two cloud elements (CE3 and CE2) exist, which then merged to form a cloud element at 12:00:00 UTC (F13CE3) (CC_1 top left portion of Fig. 3a). Splitting of CEs also occurred as seen with the cloud element at 18:00:00 UTC (F19CE3) (left middle portion of Fig. 3a). This is observed as two cloud elements at 20:00:00 UTC (F20CE1 and F20CE4). New features are also observed to have formed and become part of CCs such as with F20CE5. The graph also indicates varying durations of CCs – the shortest is 2 h while the longest is 14 h.

The MCSs search identified 6 systems (Fig. 3b). Most of these MCSs are simple, short-lived MCSs ranging between 3 and 7 h duration, with the exception of one feature that has duration of 14 h and shows a complicated lifetime from F12 through to F26 (CC_1 - left of Fig. 3b). Fig. 3c provides a comparison of the original satellite images and the identified CEs between 11 September 2006 17:00:00 UTC and 20:00:00 UTC inclusive, which corresponds to the nodes in the blue box in the abstract graph representation of the CC in Fig. 3a and the MCS in Fig. 3b). It is observed that the shape characteristics of the cloud areas are maintained, furthermore even though cloud masking at a the temperature threshold was facilitated, the temperature characteristics of the cloud are maintained i.e. the colder cloud areas are kept and thus can be further analyzed in an iterative fashion. The feature is observed as one CE at 18:00:00 UTC as F19CE2, formed from the merging of the large CE (F18CE1) and the smaller CE (F18CE2) from the previous hour. At 18:00:00 UTC, it is noted that the previous hour, the F18CE1, split into two parts – F19CE1 and F19CE2 (Fig. 3a), but the MCS feature – the core of the CC – is tracked as F19CE2 (Fig. 3b). The results indicate the ability of the algorithm to identify the temperature and shape characteristics of interest from the IR satellite data and the complexity of the system. The features identified include cloud clusters (CCs) of related cloudy areas through time, and core MCSs – the areas most closely correlated in each CC. Depending on the application of the use of the algorithm, a user may decide to further explore the graph representing the CC, e.g. for operational forecasters interested in the system in its entirety, or for researchers interested in the core system, i.e. the MCS. Further discussion on this case study is provided for the MCS representation.

A graphical representation of the area contribution of each CE to the main MCS feature is provided in Fig. 4 to illustrate an example of the visualization available within the algorithm. The CE elements associated with MCCs in West Africa should be 80,000 km^2 or larger collectively at a given time, but smaller than the global population average (O (2–5 $\times 10^5$ km^2)), according to Laing & Fritsch (1993). As such, this visualization provides an idea to the size evolution of the core feature and its maximum extent. The MCS feature reached its maximum extent at 20:00:00 UTC with a maximum area size of 131,392.0 km^2. The results from this algorithm are spatially and temporally consistent with the other algorithms in the literature used to detect this MCS.

The GTG algorithm allows for an evaluation of the precipitation characteristics of identified CCs and MCSs using TRMM data. As a preliminary visualization of this capability within the algorithm, the total rainfall associated with the MCS was determined (Fig. 5), and the amount of rainfall associated with each CE that contributed. Fig. 6 indicates the percentage of the identified cloud element (area from TRMM dataset/ area from the MERG dataset) that precipitated throughout the duration of the feature, and its location relative to the center of the feature. The initial stages of the MCS indicate that the CEs were mostly precipitating, as indicated by the large percentages. In the mature stage of the MCS, the precipitation area was greatly reduced as compared to the feature.

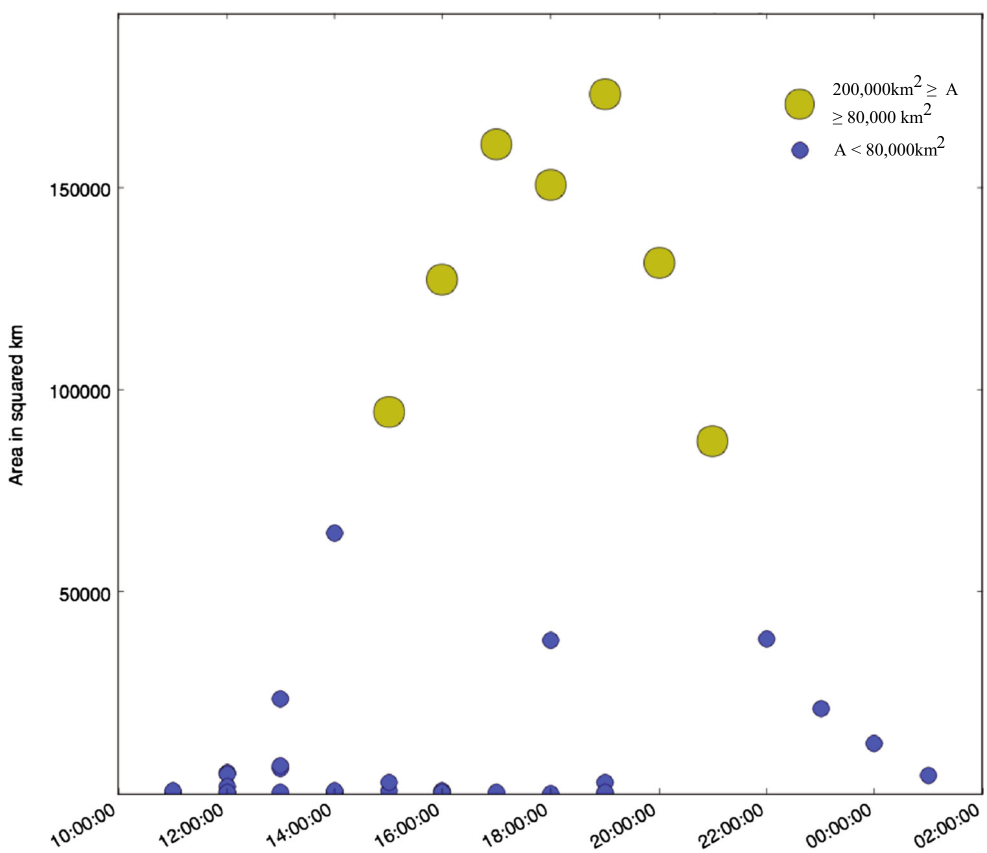

Fig. 4 The area distribution for the MCS observed over Niamey, Niger between 11 September 2006 11:00:00UTC and 12 September 2006 02:00:00UTC. The dots qualitatively represent the area of the cloud element. The large red dot represents CEs areas larger than 200,000 km^2, the yellow dot areas between 80,00 km^2 – 200,000 km^2 and the blue dot, < 80,000 km^2

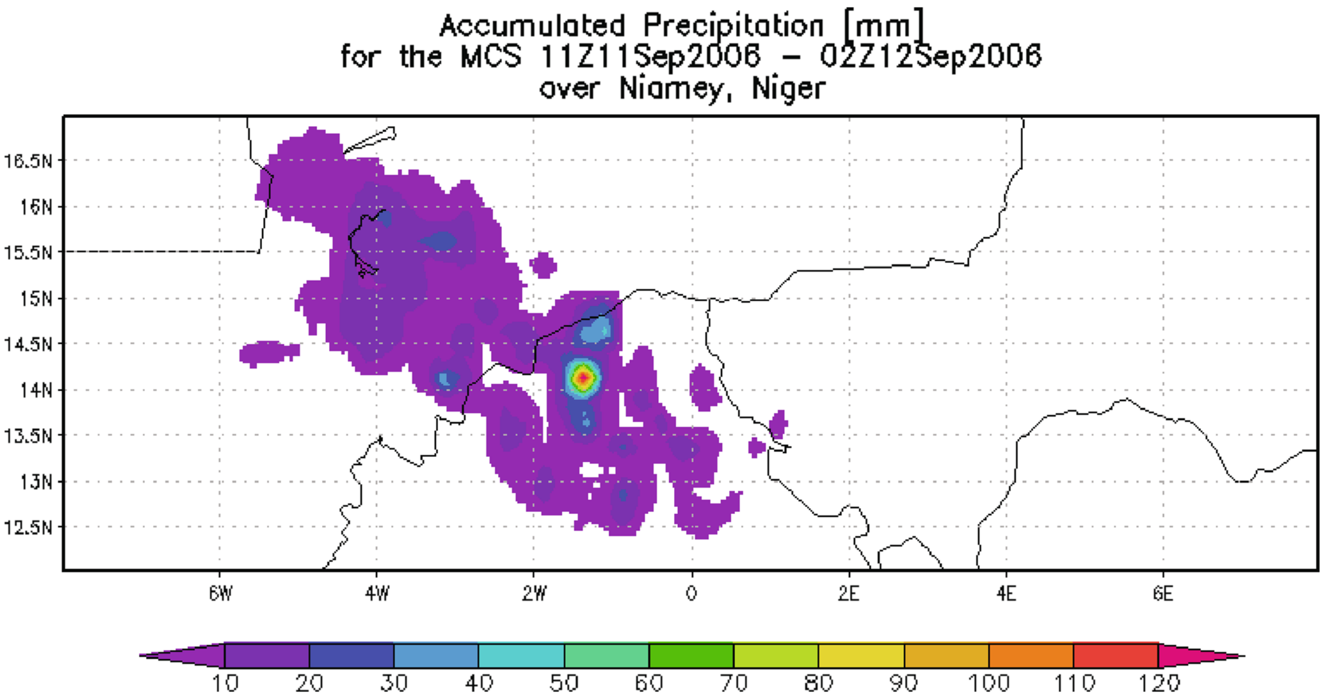

Fig. 5 The accumulated precipitation in mm for the duration of the MCS over Niamey, Niger

The MCC search did not identify any MCCs, as is consistent with the literature.

Sensitivity of the cloud shield brightness temperature to the detection of graph nodes and features

In this section, the GTG algorithm is examined for different brightness temperature thresholds. The BT parameter is changed from 241 K, to 235 K, 220 K, 205 K to identify more convectively active areas, and 250 K to identify warm cloud areas. At 250 K-BT, 91 nodes, 80 edges, and 12 CCs are identified, while at 235 K-BT 59 nodes, 54 edges, and 8 CCs are identified. At 220 K-BT, 42 nodes, 38 edges and 5 CCs are identified, and at 205 K-BT 16 nodes, 13 edges and 3 CCs. In all cases, no MCC was identified. In general, the complexity of the graphs and the duration of the CCs reduce as colder BTs are analyzed.

Sensitivity of the percentage overlap to the detection of graph edges

In this section, the GTG algorithm is examined for varying percentage overlap values for the weighted edges. The weighed edges are changed to minimum weight for 95 %

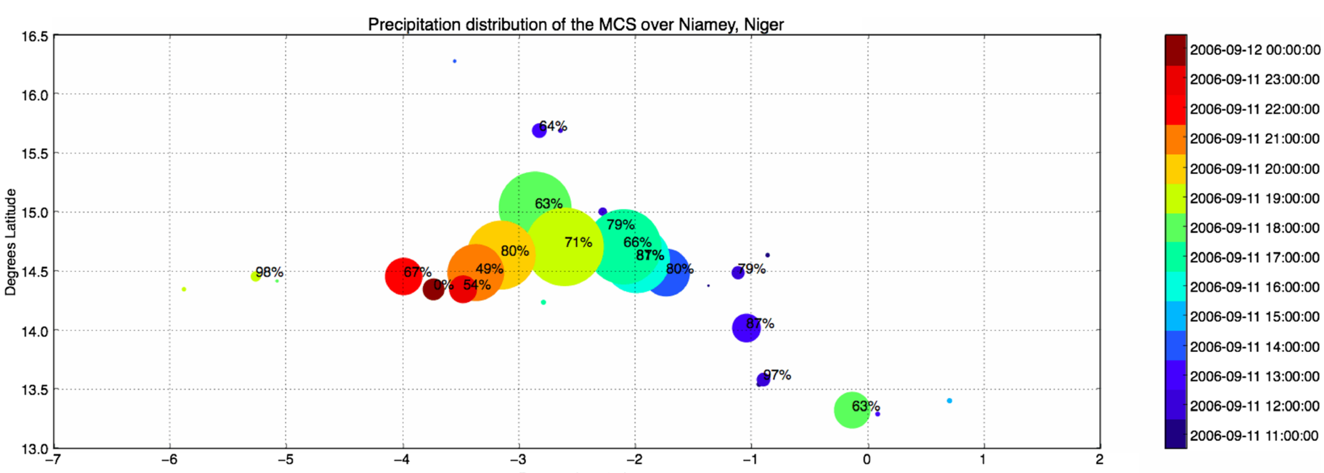

Fig. 6 The spatial and temporal distribution of precipitation of CEs≥2,400 km^2 in the MCS. The dots represent the relative area of each CE to the total area of the system. The percentage represents the percentage of the CE that was precipitating according to the TRMM data

overlap and maximum weight for 90 % - 95 % overlap. It was observed that the larger the percentage overlap requirement, the fewer small CEs – most common at the initiation and decay stages of the MCS – are captured in the evolution. This highlights the known dependency of the percentage overlap on the temporal resolution of the dataset. However, the core system (i.e. the system during the mature stage) is still captured.

Case study: Tracking a MCC over Burkina Faso

The area over Burkina Faso (5°N - 19°N, 5°E - 9°W) is examined for the 48 h period between 31 August 2009 00:00:00 UTC to 1 September 2009 23:00:00 UTC using the IR temperature data from the MERG dataset and the 3-hourly TRMMv7 precipitation rate dataset. This location and time period was chosen as we wish to compare our results with Galvin (2010) who provides statistics on the MCC feature commencement, duration, and size. In order to detect MCCs, a minimum BT of 241 K must be used as according to Maddox (1980), this is the BT of the outer shield. However, further analysis on identified CEs is possible to determine their convective structure, and/or graph structures can be constructed using different brightness temperature thresholds for comparison.

The GTG algorithm identified a dense graph with 169 nodes and 159 edges, and 15 cloud clusters. Of this graph, 11 MCSs were identified and one MCC that is shown in Fig. 7. The complex MCS-MCC feature commenced on 31 August 2009 03:00:00 UTC and had a MCC feature embedded within it that lasted for 21 h (from the point of identification at 31 August 2009 12:00:00 UTC until 1 September 2009 09:00:00 UTC). The GTG algorithm identified the shape and temperature criteria conform to that of the MCC criterion in Table 1, at 12:00:00 UTC, and the minimum duration of this area, and temperature criteria in order to be classified as a MCC, is met by 18:00:00 UTC. At 15:00:00 UTC on 31 August 2009, the feature area was 251,744 km^2. The noted

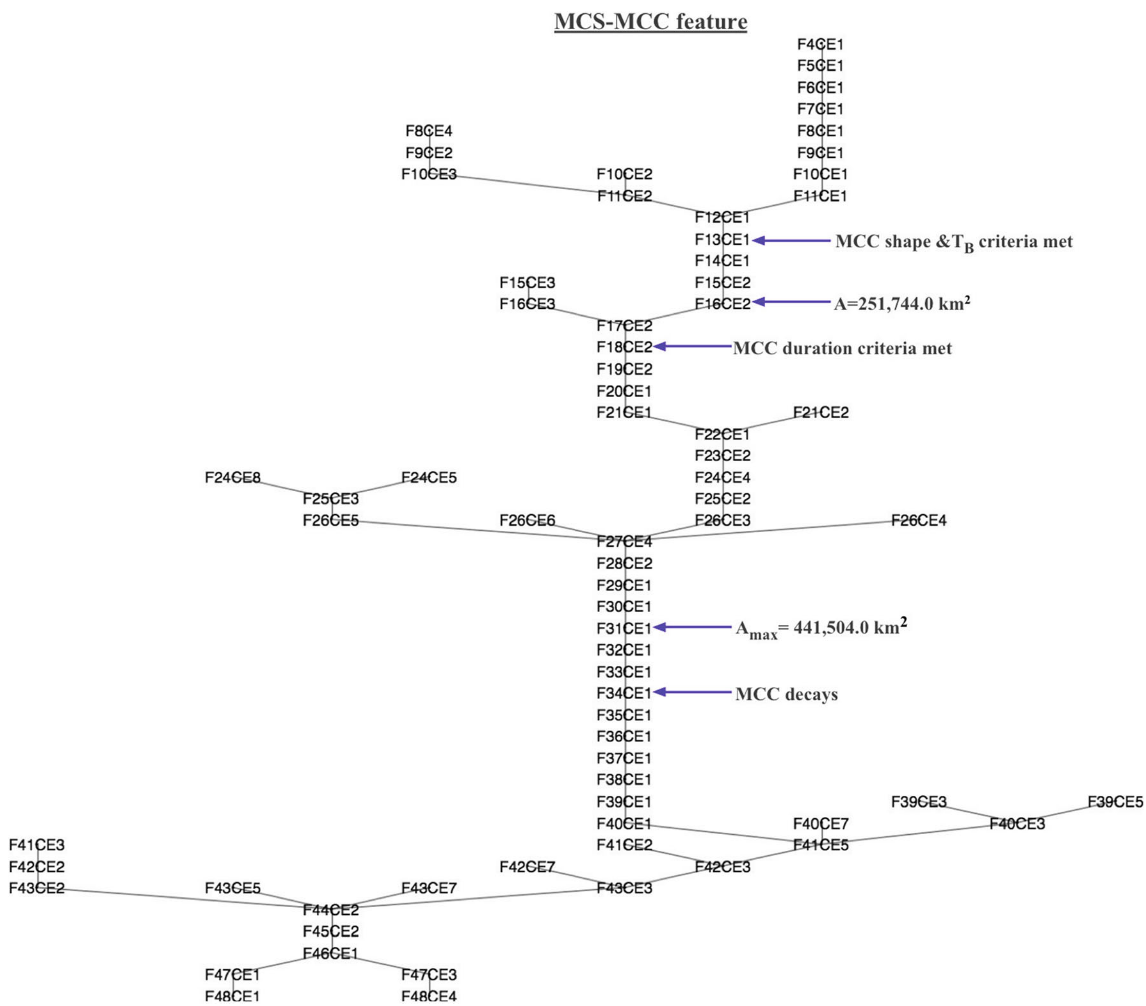

Fig. 7 The MCS-MCC feature identified over Burkina Faso during the period 31 August 2009 03:00:00 UTC and 1 September 2009 23:00:00 UTC. The text (*graph nodes*) represent the cloud elements (CEs) identified. The text indicates the frame number and the cloud element number for that frame. The black lines (*graph edges*) indicate an area overlap ≥65 % or more between CEs

initiated MCC and duration, according to the GTG algorithm correlates with the times in the literature (Galvin 2010).

Computer resourses considerations

Using the IR dataset spatial resolution, the first case study translates to 60,280 points – 440 in the x-direction and 137 in the y-direction – being analyzed at each hour, whereas the second case study contains 161, 408 spatial points – 416 in the x-direction and 388 points in the y-direction. Both case studies were implemented on a personal computer system in a fully automated end-to-end evaluation after the parameters related to the MCC criterion, domain, area and temperature thresholds were entered. The evaluations completed in a few minutes (less than 8 min each).

Conclusions and Future work

This study presented the "Grab 'em, Tag 'em, Graph 'em" (GTG) algorithm as a means a cloud-tracking and MCC feature tracking algorithm based on graph theory for implementation in gridded infrared datasets. We conclude from this study that a graph theory implementation such as the GTG algorithm can be used to (1) represent weather features from satellite data as graph; (2) inherently evolve the complex nature of these features; and (3) identify and characterize large MCS systems in one or more satellite datasets through graph searches and traversal methods. Additionally, it is concluded:

(1) The GTG algorithm efficiently uses computer resources as only the CEs are isolated from each image. Additionally, this algorithm retains all the original temperature information the delineated CE area in netCDF format for post-processing to meet individual research needs. In this way, a wider range of MCSs can be easily accounted for, by providing the appropriate method (function) to search the CCs.

(2) Brightness temperatures and weighting edges based on CEs proximity to each other spatially allow for identifying the core contributors to MCS development and aid in isolating stages of MCS growth. This distinction allows for a wider use of the algorithm in various applications.

(3) The GTG algorithm provides flexibility in choosing BT and percentage overlap parameters, and determining how to track features. For some applications it may be useful to track what is defined as the core system where as for others, the entire MCS may be needed (i.e. don't do the Djikstra's shortest path). Similarly in identifying a weather feature, e.g. MCC, some applications may require the core feature be captured, whereas others may wish to consider the embedding MCS.

From our initial studies, we are satisfied that this approach is viable and we intend to further test the method with longer durations, other datasets (for infrared and rainfall data) as well as other locations. We also anticipate developing visualization products for the outputs from the long-term record. Furthermore, we recognize the graphing method lends to use in graph databases and parallelization, and thus larger datasets (both in time and space) can be considered. As such, we hope that this method can in time address the deficiencies associated with the lack of understanding of the variability of different MCSs in the rainy season over years.

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